Synoptic Question 1: The Physics of Magnetic Levitation (Maglev) Trains

Reading Passage

Read the following article and then answer the questions that follow.

Magnetic levitation (Maglev) trains represent a significant advancement in high-speed ground transportation, utilizing magnetic forces for both levitation and propulsion. This technology eliminates the friction between wheels and rails, allowing for much higher speeds and a smoother ride compared to conventional trains. One of the most well-known examples is the Shanghai Transrapid, which reaches a top speed of 431 km/h.

The levitation is achieved through the use of powerful electromagnets. In one common design, electromagnets on the train's undercarriage are attracted upwards towards a ferromagnetic guideway. A sophisticated control system continuously adjusts the current in the electromagnets to maintain a stable levitation gap of about 10 mm.

Propulsion is provided by a linear synchronous motor. Coils in the guideway are energized in a sequence, creating a moving magnetic field that interacts with strong permanent magnets on the train. This interaction produces a force that propels the train forward. The speed of the train is controlled by the frequency of the alternating current supplied to the guideway coils.

While Maglev trains are highly energy-efficient at high speeds, they consume a significant amount of power, especially during acceleration and to maintain the levitation. The Shanghai Transrapid has a mass of 225,000 kg and operates on a 30 km track. During its normal operation, it accelerates to its top speed of 431 km/h in just 3 minutes and 30 seconds.

The main electromagnets for levitation are made from a superconducting material, which needs to be cooled to a very low temperature. The coils for the propulsion system in the guideway, however, are made of copper and have a certain electrical resistance.

Data for a typical Maglev train:

- Total mass of train, M: 225,000 kg
- Levitation gap: 10 mm
- Cruising speed, v: 431 km/h
- Acceleration time: 3 minutes 30 seconds
- Magnetic field strength for levitation, B: 0.85 T
- Total area of levitation electromagnets, A: 25 m^2
- Resistance of a single guideway propulsion coil, R: 0.20 Ω
- Number of propulsion coils energized at any one time: 150

Questions

(a) Convert the cruising speed of 431 km/h to m/s.

(b) Assuming constant acceleration, calculate:

  (i) The acceleration of the train.

  (ii) The change in momentum of the train during the acceleration phase.

(c) The levitation force must be equal to the weight of the train. The magnetic pressure P exerted by an electromagnet is given by the formula $P = \frac{B^2}{2\mu_0}$ where μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ T $\cdot$ m/A).

  (i) Calculate the total upward magnetic force supporting the train and show that it is sufficient to levitate the train.

  (ii) Discuss why a control system is necessary to maintain the 10 mm levitation gap.

(d) The kinetic energy of the train at cruising speed is provided by the linear synchronous motor.

   (i) Calculate the kinetic energy of the train at its cruising speed.

   (ii) Ignoring air resistance, calculate the average power supplied by the propulsion system during the acceleration phase.

(e) The current flowing through each of the 150 energized propulsion coils is 500 A.

   (i) Calculate the total power dissipated as heat in the guideway coils during propulsion.

   (ii) Compare the total energy dissipated as heat during the acceleration phase with the final kinetic energy of the train.

(f) The train enters a curved section of the track with a radius of 2.0 km while maintaining its cruising speed.

   (i) Calculate the centripetal acceleration of the train.

   (ii) Calculate the horizontal force that the guideway must exert on the train to keep it on the curved track.

(g) To slow down, the train can use electromagnetic braking. By reversing the direction of the moving magnetic field in the guideway, it can be made to travel slower than the train.

   (i) Explain how this creates a braking force on the train.

   (ii) State the main energy transformation that occurs during this type of braking and suggest one advantage of this system.

(h) Suggest why superconducting materials are used for the levitation magnets, despite the complex and costly cooling systems required to keep them at very low temperatures.

Solutions

(a) $v = 431 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 119.7 \text{ m/s}$

(b) (i) $t = 3 \text{ min}30 \text{ s} = 210 \text{ s}$.
$a = \frac{v-u}{t} = \frac{119.7-0}{210} = 0.570 \text{ m/s}^2$

(ii) Change in momentum, $\Delta p = m(v-u) = (225000)(119.7-0) = 2.69 \times 10^7$ kg m/s.

(c) (i) Weight of train, $W = mg = (225000)(9.81) = 2.21 \times 10^6$ N.
Magnetic Pressure, $P = \frac{B^2}{2\mu_0} = \frac{(0.85)^2}{2(4\pi \times 10^{-7})} = 2.87 \times 10^5$ Pa.
Total upward magnetic force, $F = P \times A = (2.87 \times 10^5) \times 25 = 7.18 \times 10^6$ N. Since $7.18 \times 10^6 \text{ N} > 2.21 \times 10^6$ N, the force is sufficient.

(ii) The attractive magnetic force is inherently unstable. If the gap decreases, the magnetic force increases ($F \propto 1/r^2$), pulling the train even closer. If the gap increases, the force weakens, causing it to fall. A control system is needed to continuously adjust the electromagnet current to counteract these small changes and maintain a stable gap.

(d) (i) Kinetic Energy, $KE = \frac{1}{2}Mv^2 = \frac{1}{2}(225000)(119.7)^2 = 1.61 \times 10^9$ J.

(ii) Average Power, $P_{avg} = \frac{\text{Work Done}}{\text{time}} = \frac{KE}{t} = \frac{1.61 \times 10^9}{210} = 7.67 \times 10^6 \text{ W} = 7.67 \text{ MW}$.

(e) (i) Power dissipated in one coil, $P_1 = I^2R = (500)^2(0.20) = 50000$ W.
Total power dissipated, $P_{total} = 150 \times 50000 = 7.5 \times 10^6 \text{ W} = 7.5 \text{ MW}$.

(ii) Energy dissipated as heat $= P_{total} \times t = (7.5 \times 10^6) \times 210 = 1.58 \times 10^9$ J. The energy lost as heat (1.58×10^9 J) is very similar in magnitude to the final kinetic energy of the train (1.61×10^9 J), showing that a large amount of energy is wasted overcoming the resistance of the coils.

(f) (i) Centripetal acceleration, $a_c = \frac{v^2}{r} = \frac{(119.7)^2}{2000} = 7.16 \text{ m/s}^2$.

(ii) Horizontal force, $F_c = Ma_c = (225000)(7.16) = 1.61 \times 10^6$ N.

(g) (i) The magnets on the train are moving faster than the magnetic field from the guideway. This relative motion induces eddy currents in the guideway coils. According to Lenz's law, these induced currents create a magnetic field that opposes the motion that caused them, resulting in a braking (retarding) force on the train.

(ii) The main energy transformation is from the train's **kinetic energy** into **electrical energy** in the guideway coils. The advantage of this **regenerative braking** is that this electrical energy can be returned to the power grid, making the system more energy-efficient.

(h) Superconducting materials have zero electrical resistance at low temperatures. Using them for the levitation magnets means that once the current is established, it can flow indefinitely without any loss of energy as heat. This makes it extremely energy-efficient to maintain the strong magnetic field required for levitation continuously.

Figure 1: Diagram of Maglev Train Propulsion

Synoptic Question (CIE P4 Style) – Variant 2 (Higher Difficulty)

Reading Passage

Magnetic levitation (Maglev) systems remove wheel–rail contact and can therefore reach very high speeds. In a high-speed test section, a train of total mass 2.25×10^5 kg accelerates from rest to $120\,\mathrm{m\,s^{-1}}$ in 180 s. During this stage, the average traction force is produced by a linear synchronous motor interacting with guideway coils.

At cruising speed, the train enters a banked curve of radius 1.80×10^3 m. The guideway is designed so that, at one specific speed, no lateral magnetic correction is needed.

The levitation module can be modeled as producing magnetic pressure

$$P = \frac{B^2}{2\mu_0},$$

where B is magnetic flux density and $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$. The effective levitation area is $24\,\mathrm{m}^2$ and the nominal field is 0.72 T.

Each active propulsion coil has resistance $0.18\,\Omega$ and carries current 420 A. At a given instant, 180 coils are energized.

To improve sustainability, the system uses regenerative braking, but only part of the recovered electrical energy can be returned to the grid due to conversion losses.

Comprehension Extract (read and extract data)

A prototype Maglev set with mass 2.25×10^5 kg starts from rest and reaches $120\,\mathrm{m\,s^{-1}}$ in 180 s. It then enters a curve of radius 1.80×10^3 m. The levitation system operates near $B = 0.72$ T across an effective area of $24\,\mathrm{m}^2$, with $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$. In propulsion mode, 180 coils are energized; each has resistance $0.18\,\Omega$ and carries current 420 A. Take $g = 9.81\,\mathrm{m\,s^{-2}}$. During braking, assume 70% of kinetic energy is first converted to electrical energy and then only 60% of that is exported to the grid.

Questions

(a) Using data from the comprehension extract, determine the train's acceleration and the net horizontal force during the acceleration phase.

(b) Calculate the train's kinetic energy at top speed and hence the average mechanical power delivered to the train over the acceleration interval.

(c) For the curved section:

- (i) calculate the required centripetal acceleration,
- (ii) calculate the corresponding centripetal force,
- (iii) show that the design bank angle for zero lateral correction satisfies $\tan\theta = v^2/(rg)$ and find θ.

(d) Using magnetic pressure, calculate the total upward magnetic force available from the levitation system and compare this with the train's weight. Comment on the implication for operation.

(e) Calculate total Joule-heating power in the energized propulsion coils.

(f) The train brakes from top speed to rest.

  (i) Find the total energy returned to the grid.

  (ii) State one reason why 100% recovery is impossible.

(g) **Open-ended conceptual response (higher difficulty):**

> A proposal is made to increase cruising speed further while keeping passenger comfort, energy efficiency, and safety acceptable. Discuss, using relevant physics principles, how at least **three** of the following are affected and what design changes would be required: curve dynamics, levitation stability, electromagnetic heating, and regenerative braking effectiveness.

Your answer should include linked arguments, not isolated statements.

Sample Solution (Variant 2)

(a)

$$a = \frac{v-u}{t} = \frac{120-0}{180} = 0.667\,\mathrm{m\,s^{-2}}$$

$$F = Ma = (2.25\times10^5)(0.667) = 1.50\times10^5\,\mathrm{N}$$

(b)

$$E_k = \frac{1}{2}Mv^2 = \frac{1}{2}(2.25\times10^5)(120)^2 = 1.62\times10^9\,\mathrm{J}$$

$$P_{\mathrm{avg}} = \frac{E_k}{t} = \frac{1.62\times10^9}{180} = 9.00\times10^6\,\mathrm{W} = 9.00\,\mathrm{MW}$$

(c) (i) $a_c = v^2/r = 120^2/(1.80\times10^3) = 8.00\,\mathrm{m\,s^{-2}}$

(ii) $F_c = Ma_c = (2.25\times10^5)(8.00) = 1.80\times10^6\,\mathrm{N}$

(iii) For a banked track with no lateral correction:

$$\tan\theta = \frac{v^2}{rg} = \frac{120^2}{(1.80\times10^3)(9.81)} = 0.815$$

$$\theta = \tan^{-1}(0.815) = 39.2^\circ$$

(d)

$$P = \frac{B^2}{2\mu_0} = \frac{0.72^2}{2(4\pi\times10^{-7})} = 2.06\times10^5\,\mathrm{Pa}$$

$$F_{\mathrm{mag}} = PA = (2.06\times10^5)(24) = 4.95\times10^6\,\mathrm{N}$$

$$W = Mg = (2.25\times10^5)(9.81) = 2.21\times10^6\,\mathrm{N}$$

Since $F_{\mathrm{mag}} > W$, levitation is feasible with margin. In practice, active control is still needed for gap stability and disturbance rejection.

(e) Power per coil:

$$P_1 = I^2R = (420)^2(0.18) = 3.1752\times10^4\,\mathrm{W}$$

Total:

$$P_{\mathrm{heat}} = nP_1 = 180(3.1752\times10^4) = 5.72\times10^6\,\mathrm{W} = 5.72\,\mathrm{MW}$$

(f) (i) Initial kinetic energy is 1.62×10^9 J. Electrical conversion: $0.70E_k$. Grid return: $0.60(0.70E_k) = 0.42E_k$.

$$E_{\mathrm{grid}} = 0.42(1.62\times10^9) = 6.80\times10^8\,\mathrm{J}$$

(ii) Any valid reason, e.g. resistive losses in coils/cables, power electronics inefficiency, magnetic hysteresis and eddy-current losses, or control constraints to preserve ride comfort.

(g) **Indicative high-level response:** Increasing speed raises required centripetal acceleration as $a_c \propto v^2$, so tighter curves become less acceptable unless radius or bank angle increases. Higher speed also increases propulsion demand and typically increases electromagnetic losses because larger currents and switching frequencies are needed. Levitation stability becomes more challenging because dynamic disturbances and control bandwidth requirements rise; faster feedback and robust control are needed to keep the air gap stable. Regenerative braking power potential increases at high speed (more kinetic energy), but thermal and converter limits can cap real recovery. Therefore, design upgrades must combine larger-radius alignments, optimized banking, improved cooling/conductors, and higher-rated power electronics with advanced control.

Mark Scheme (CIE P4 Style) – Variant 2 Only

Total: 40 marks

Q	Answer points	Marks
(a)	$a = 0.667\,\mathrm{m\,s^{-2}}$ (allow 0.66–0.67); method using $a = (v-u)/t$	B1 M1
	$F = 1.50 \times 10^5\,\mathrm{N}$ with use of $F = Ma$	B1
(b)	Correct substitution in $E_k = \frac{1}{2}Mv^2$	M1
	$E_k = 1.62 \times 10^9\,\mathrm{J}$	A1
	Correct use of $P = E/t$	M1
	$P_{\mathrm{avg}} = 9.0 \times 10^6\,\mathrm{W}$ (9.0 MW)	A1
(c)(i)	$a_c = 8.00\,\mathrm{m\,s^{-2}}$ via v^2/r	M1 A1
(c)(ii)	$F_c = 1.80 \times 10^6\,\mathrm{N}$ via Ma_c	M1 A1
(c)(iii)	States/uses $\tan\theta = v^2/(rg)$	B1
	Substitutes values consistently	M1
	$\theta \approx 39^\circ$ (38.5–40.0 accepted)	A1
(d)	Correct expression for magnetic pressure	B1
	$P \approx 2.06 \times 10^5\,\mathrm{Pa}$	A1
	Uses $F = PA$ and gets $F_{\mathrm{mag}} \approx 4.95 \times 10^6\,\mathrm{N}$	M1 A1
	Calculates weight $W \approx 2.21 \times 10^6\,\mathrm{N}$ and valid comparison/comment	B1 A1
(e)	Uses $P_1 = I^2R$ correctly for one coil	M1
	$P_1 \approx 3.18 \times 10^4\,\mathrm{W}$	A1
	Multiplies by 180 and gets $P_{\mathrm{heat}} \approx 5.72 \times 10^6\,\mathrm{W}$	M1 A1
(f)(i)	Uses two efficiency factors multiplicatively (0.70 and 0.60)	M1
	$E_{\mathrm{grid}} = 6.8 \times 10^8\,\mathrm{J}$	A1
(f)(ii)	One valid physical limitation stated	B1
(g)	Up to 12 marks, level of response (open-ended): **L3 (9–12):** coherent, linked physics argument covering at least three listed domains; explicit use of scaling ideas (e.g., v^2 effects), realistic engineering trade-offs, and justified design modifications. **L2 (5–8):** some correct multi-domain discussion with partial linkage; may contain minor omissions or weak justification. **L1 (1–4):** isolated statements, limited physics depth, little or no linkage between concepts. **0:** no creditable response.	